

Algorithms Problem Solving

Level 2

My SQRT

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🔍 Definition

- ◆ This program calculates the square root of a number without directly using the built-in `sqrt()` function.
- ◆ The square root of a number is a value that, when multiplied by itself, gives the original number.
- ◆ The program uses `pow(Number, 0.5)` to calculate the square root.
- ◆ The result is returned as a `short`, so the decimal part is removed.

Explanation

- ◆ ``ReadNumber()`` reads a number from the user as a ``float``.
- ◆ ``MySqrt()`` receives the number and calculates:
 $\text{pow}(\text{Number}, 0.5)$
- ◆ ``0.5`` is equivalent to ``1 / 2``, so:
 $\text{Number}^{0.5} = \sqrt{\text{Number}}$
- ◆ For example:
 $\sqrt{226} \approx 15.0333$
- ◆ Because ``MySqrt()`` returns a ``short``, the decimal part is discarded:
 $15.0333 \rightarrow 15$
- ◆ The built-in ``sqrt()`` returns the more precise floating-point result.

☰ Steps in main()

- 1☐ Ask the user to enter a number.
- 2☐ Pass the number to `MySqrt()`.
- 3☐ Calculate the square root using `pow(Number, 0.5)`.
- 4☐ Return the result as a `short`.
- 5☐ Print the result from `MySqrt()`.
- 6☐ Calculate the square root using C++'s built-in `sqrt()`.
- 7☐ Print the built-in result for comparison.

💡 Example

◆ Input:

226

◆ Calculate:

$\text{pow}(226, 0.5) \approx 15.0333$

◆ Since `MySqrt()` returns `short`:

$15.0333 \rightarrow 15$

◆ The built-in `sqrt()` keeps the decimal part:

$\text{sqrt}(226) \approx 15.0333$

➡ Result

My Sqrt Result: 15

C++ Sqrt Result: 15.0333

✈ Note

- ◆ Important function:

```
short MySqrt(short Number)
{
    return pow(Number, 0.5);
}
```

- ◆ The function uses `pow()` instead of `sqrt()`.
- ◆ `pow(Number, 0.5)` mathematically represents the square root.
- ◆ The return type `short` causes the decimal part to be truncated.
- ◆ There is a type mismatch in the program:
float number
is passed to:
short MySqrt(short Number)
- ◆ Therefore, the `float` input is converted to `short` before the function calculates the square root.
- ◆ A better version would use `float` or `double` for the parameter and return type if you want to preserve the decimal result.

⊘ Warning

- ◆ The program does not actually implement the square-root calculation from scratch; it uses ``pow()``, which is still a built-in mathematical function.
- ◆ If the exercise means "do not use any built-in square-root-related function", then ``pow()`` would also not satisfy that requirement.
- ◆ ``MySqrt()`` only accepts a ``short``, even though the user input is a ``float``.
- ◆ Negative numbers do not have a real square root.
- ◆ ``corecrt_math_defines.h`` is not needed.
- ◆ ``cmath`` is needed for ``pow()`` and ``sqrt()``.

Comparison

`MySqrt()`

- ✓ User-defined function.
- ✓ Uses `pow(Number, 0.5)` to calculate the square root.
- ✓ Returns a `short`, so decimal values are discarded.
- ✓ Example: $\sqrt{226} \approx 15.0333 \rightarrow 15$.

`sqrt()`

- ✓ Built-in C++ function.
- ✓ Directly calculates the square root.
- ✓ Keeps the decimal part.
- ✓ Example: $\sqrt{226} \approx 15.0333$.

✦ Summary

- ✓ Read a number from the user.
- ✓ Calculate its square root using `pow(Number, 0.5)`.
- ✓ Return the result as a `short`.
- ✓ The `short` return type removes the decimal part.
- ✓ Compare the result with C++'s `sqrt()` function.
- ✓ For `226`, `MySqrt()` returns `15`, while `sqrt()` returns approximately `15.0333`.

Thank you 😊